



ADDIS ABABA UNIVERSITY
COLLEGE OF NATURAL AND COMPUTATIONAL
SCIENCES
DEPARTMENT OF MICROBIAL SCIENCES AND
GENETICS
MASTERS OF SCIENCE IN BIOINFORMATICS

Advanced programing for Bioinformatics
Assignment-1

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Task 1: Environment Setup

Python is a widely used programming language in bioinformatics due to its simplicity and versatility. It supports many scientific libraries such as NumPy and Biopython, which are useful for data analysis and biological computations. Because of its readable syntax, it is suitable for both beginners and advanced users.

Perl is commonly used in bioinformatics for processing and analyzing text-based data, especially biological sequence files. It is very efficient when working with large datasets and performing pattern matching. Many existing bioinformatics tools and scripts are written in Perl.

R is a statistical programming language mainly used for data analysis and visualization. In bioinformatics, it is particularly helpful for analyzing experimental data such as gene expression. Packages like ggplot2 and Bioconductor make it powerful for research purposes.

A text editor is necessary for writing and editing code. Visual Studio Code is a popular choice because it offers useful features like syntax highlighting, extensions, and debugging tools. Simpler editors such as nano and vim can also be used, especially when working in a terminal environment.

```
Last login: Thu Apr 23 13:19:55 on ttys001
(base) dejene@Dejenes-MacBook-Air ~ % python --version
Python 3.12.7
(base) dejene@Dejenes-MacBook-Air ~ % perl -v

This is perl 5, version 42, subversion 2 (v5.42.2) built for darwin-thread-multi-2level
Copyright 1987-2026, Larry Wall

Perl may be copied only under the terms of either the Artistic License or the
GNU General Public License, which may be found in the Perl 5 source kit.

Complete documentation for Perl, including FAQ lists, should be found on
this system using "man perl" or "perldoc perl". If you have access to the
Internet, point your browser at https://www.perl.org/, the Perl Home Page.

(base) dejene@Dejenes-MacBook-Air ~ % R --version
R version 4.5.3 (2026-03-11) -- "Reassured Reassurer"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: aarch64-apple-darwin25.3.0

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under the terms of the
GNU General Public License versions 2 or 3.
For more information about these matters see
https://www.gnu.org/licenses/.

(base) dejene@Dejenes-MacBook-Air ~ % code --version
1.96.2
fabdb6a30b49f79a7aba0f2ad9df9b399473380f
arm64
(base) dejene@Dejenes-MacBook-Air ~ %
```

Task 2: Linux File System Navigation

```
(base) dejene@Dejenes-MacBook-Air ~ % pwd
/Users/dejene
(base) dejene@Dejenes-MacBook-Air ~ % ls
3eml.cif      bioinfo      cleaned_3EML.pse  day4.ipynb      Downloads      Movies      Pictures      script.sh
adenosine.pdb bioinfo_lab  data.log          day5.ipynb      file.txt       Music      Public        workflow_output
anaconda_projects  caffeine.pdb  day2.ipynb       Desktop         iCloud Drive (Archive)  myenv       PycharmProjects  Zotero
Applications      capsaicin.pdb day3.ipynb       Documents       Library        OR0        scikit_learn_data

(base) dejene@Dejenes-MacBook-Air ~ % mkdir bioinfo_lab1
(base) dejene@Dejenes-MacBook-Air ~ % cd bioinfo_lab1
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % touch sample.txt
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % cp sample.txt copy.txt
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % mv copy.txt renamed.txt
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 %
```

1. What does **pwd** do?

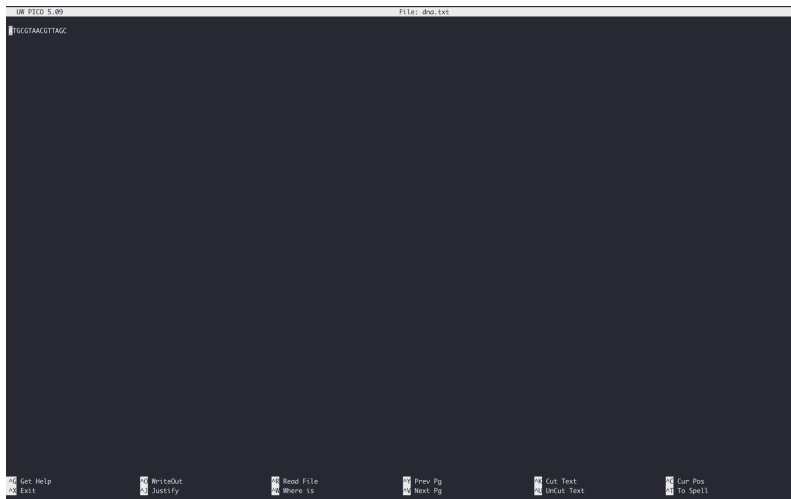
The **pwd** (print working directory) command displays the full path of the directory you are currently working in. It helps users know their exact location in the Linux file system.

2. Difference between **cp** and **mv**

- **cp (copy):**
Creates a duplicate of a file or directory. The original file remains unchanged.
- **mv (move):**
Moves a file to another location or renames it. The original file name or location is changed.

Task 3: File Handling & Text Processing

```
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % mv copy.txt renamed.txt
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % touch dna.txt
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % nano dna.txt
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % cat dna.txt
ATGCGTAACGTTAGC
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % grep "CGT" dna.txt
ATGCGTAACGTTAGC
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % grep "CGT" dna.txt
ATGCGTAACGTTAGC
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % grep "CGT" dna.txt
ATGCGTAACGTTAGC
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % echo "ATGCGTAACGTTAGC" > dna.txt
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % grep "CGT" dna.txt
ATGCGTAACGTTAGC
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % grep --color=auto "CGT" dna.txt
ATGCGTAACGTTAGC
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % nano dna.txt
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 %
```



1. What does **grep** do?

grep is a command-line tool used to search for specific patterns or text inside files. It reads each line of a file and prints only the lines that match the given pattern. In bioinformatics, it is commonly used to search for DNA or protein motifs within large sequence datasets.

2. What is the output of the search?

The command:

```
grep "CGT" dna.txt
```

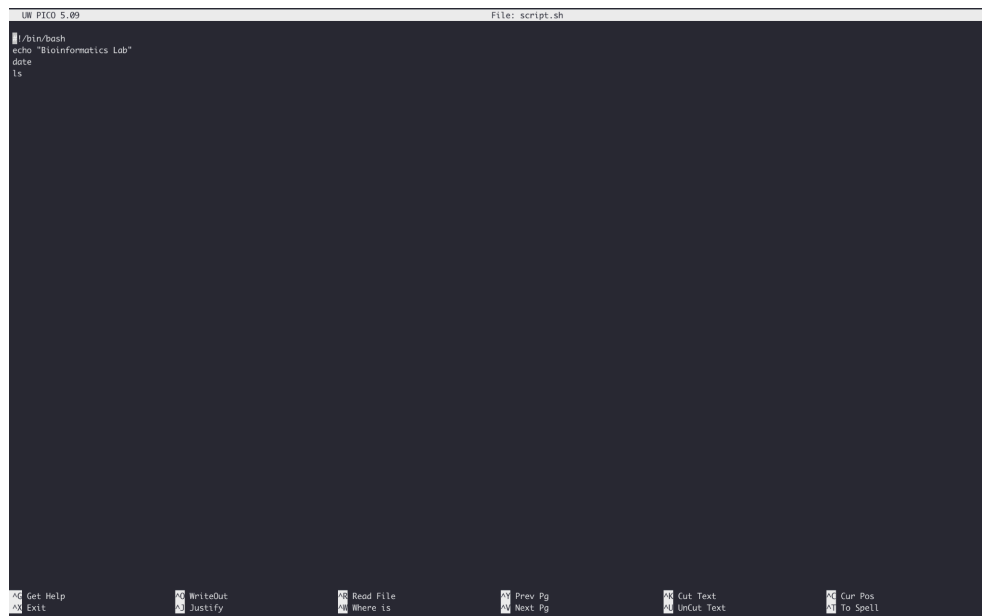
returns the line:

```
ATGCGTAACGTTAGC
```

because the sequence contains the pattern **CGT**.

Task 4: Shell Scripting

```
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % nano script.sh
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % chmod +x script.sh
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 % ./script.sh
Bioinformatics Lab
Fri Apr 24 12:50:34 EAT 2026
dna.txt      renamed.txt  sample.txt   script.sh    script.txt
(base) dejene@Dejenes-MacBook-Air bioinfo_lab1 %
```

A screenshot of a terminal window titled "UW PICO 5.09" and "File: script.sh". The terminal shows the execution of a script with the following commands and output:

```
#!/bin/bash
echo "Bioinformatics Lab"
date
ls
```

 The output of the script is visible in the terminal. At the bottom of the terminal window, there is a status bar with various icons and text: "Get Help", "Exit", "WriteOut", "Justify", "Read File", "Where is", "Prev Pg", "Next Pg", "Cut Text", "UnCut Text", "Cur Pos", and "To Spell".

1. `#!/bin/bash`

This is called a **shebang**. It tells the system to use the Bash shell to execute the script.

2. `echo "Bioinformatics Lab"`

This prints the text **Bioinformatics Lab** to the terminal.

3. `date`

Displays the current system date and time.

4. `ls`

Lists all files and directories in the current working directory.

5. `chmod +x script.sh`

This command changes file permissions and makes the script executable so it can be run directly.

6. `./script.sh`

Runs the script from the current directory.

THANK YOU!!!!

